- **Building Energy Consumption Forecasting: A Comparison of Gradient Boosting Models**

Link - Building Energy Consumption Forecasting: A Comparison of Gradient Boosting Models | Proceedings of the 12th International Conference on Advances in Information Technology

Method, Dataset Size, Key Metric, Strength, Weakness

Method:
The study applies three gradient boosting algorithms—**LightGBM, XGBoost, and CatBoost**—to compare their performance in building energy consumption forecasting.

Dataset Size:
The models are trained on a **synthesized dataset representing a large office building in Chicago**, including energy consumption and environmental variables. However, the exact dataset size is not explicitly specified in the paper.

Key Metric:
Model performance is evaluated through **prediction accuracy using error-based comparison metrics (e.g., RMSE-type evaluation)** presented in the results section.

Strength:
The paper provides a **clear comparative evaluation of advanced boosting models**, demonstrating their effectiveness in capturing nonlinear relationships in building energy data.

Weakness:
The study relies on a **single simulated building dataset**, which may limit generalisability to diverse real-world building environments.

Overlap with Our Approach

The paper strongly overlaps with our approach in the following ways:

- Uses **gradient boosting techniques (LightGBM, CatBoost, XGBoost)** — same core modelling family as our project
- Works on **structured tabular data (energy + features)** — same as ASHRAE dataset
- Relies on **feature-based learning**, where input variables (weather, building data) drive prediction

Justification for CatBoost over Deep Learning

Although deep learning is widely used in energy prediction, this paper demonstrates that tree-based gradient boosting models such as CatBoost can achieve strong predictive performance without requiring large-scale deep architectures.

This supports our choice of CatBoost because:

- It handles tabular data more effectively than deep learning
- Requires less computational cost and tuning
- Performs well even on moderate-sized datasets

Where found in paper:
Method: Methodology section
Dataset: Data description section
Metric: Results section

- **Forecasting the total building energy based on its architectural features using a combination of CatBoost and meta-heuristic algorithms**

Link - Forecasting the total building energy based on its architectural features using a combination of CatBoost and meta-heuristic algorithms - Xiaoyu Qu, Ziheng Liu, 2026

Method, Dataset, Metric, Strength, Weakness

Method:
The study proposes a hybrid machine learning approach combining **CatBoost with multiple meta-heuristic optimisation algorithms** (e.g., PSO, SSA, GWO) to improve prediction accuracy.

Dataset Size:
The model is applied to a **residential building energy dataset based on architectural features**, focusing on predicting total building energy consumption.

Key Metric:
Model performance is evaluated using **prediction accuracy improvements through optimisation-based comparison (error reduction / accuracy improvement)**.

Strength:
A key strength is the **integration of CatBoost with optimisation algorithms**, which significantly enhances predictive performance by tuning hyperparameters effectively.

Weakness:
The model introduces **high computational complexity** due to multiple meta-heuristic optimisation techniques, making it less efficient for real-time applications.

Overlap with our Approach

This paper strongly aligns with our approach in the following ways:

- Uses **CatBoost (gradient boosting model)** → same core method
- Works on **tabular structured data (building + features)**
- Relies on **feature-based modelling (architectural inputs → energy output)**

Justification for CatBoost over Deep Learning

This paper strongly justifies the use of CatBoost over deep learning because:

- It highlights that **CatBoost can achieve high predictive accuracy through optimisation without requiring deep neural networks**
- Works efficiently on **tabular feature-based datasets**, where deep learning is often less effective

- Avoids **complex architecture and large data requirements of deep learning models.**

Where found in paper:
Method: Model section
Dataset: Dataset section
Metric: Evaluation section

➢ **Building Energy Consumption Prediction Using CatBoost With Hybrid Random Search and Bayesian Optimization**

Link - Building Energy Consumption Prediction Using CatBoost With Hybrid Random Search and Bayesian Optimization | IEEE Journals & Magazine | IEEE Xplore

Method, Dataset Size, Key Metric, Strength, Weakness

Method:
The study proposes a **CatBoost-based regression model enhanced with hybrid hyperparameter optimisation techniques**, combining Random Search and Bayesian Optimisation to improve predictive performance.

Dataset Size:
The model is trained on a **building energy dataset for heating and cooling load prediction**, commonly derived from benchmark datasets such as the UCI Energy Efficiency dataset (with several hundred instances and multiple building-related features). *(Exact size may vary depending on version used in study)*

Key Metric:
Model performance is evaluated using **Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 (coefficient of determination)**.

Strength:
A key strength of the study is that **CatBoost, combined with hybrid optimisation, achieves very high prediction accuracy ($R^2 \approx 0.99$)**, outperforming multiple baseline machine learning models.

Weakness:
A limitation is that the model is **evaluated on a relatively standard benchmark dataset**, which may limit its generalisability to more complex real-world building scenarios.

Overlap with Our Approach

This paper strongly overlaps with our approach in the following ways:

- Uses **CatBoost (gradient boosting model)** → same core algorithm
- Works on **structured tabular building data (features → energy output)**
- Involves **feature-based modelling**, where building characteristics are used as inputs
- Includes **model optimisation and preprocessing**, aligning with our pipeline

Justification for CatBoost over Deep Learning

The paper supports the use of CatBoost over deep learning in the following ways:

- Demonstrates that **CatBoost achieves extremely high accuracy without requiring deep neural networks**
- Works efficiently on **tabular datasets**, where deep learning models are often less effective
- Requires **lower computational cost and simpler architecture compared to deep learning models**
- Shows that **optimised boosting models can outperform or match complex models in structured data tasks**

Where found in paper:
Method: Methodology
Dataset: Dataset description
Metric: Results

➢ **Interpretable building energy performance prediction using XGBoost Quantile Regression**

Link - [Interpretable building energy performance prediction using XGBoost Quantile Regression - ScienceDirect]

Method, Dataset, Metric, Strength, Weakness)

Method:
The study applies multiple machine learning models for building energy prediction, including Artificial Neural Networks (ANN), Support Vector Machines (SVM), Random Forest (RF), and **Extreme Gradient Boosting (XGBoost)**. The final model is enhanced using SHAP (SHapley Additive Explanations) to improve interpretability and understand feature importance.

Dataset Size:
The model is trained on a real-world dataset consisting of **3,594 residential building samples**, containing building characteristics and energy-related variables.

Key Metric:
Model performance is evaluated using standard regression metrics such as:

- Root Mean Squared Error (**RMSE**)
- Mean Absolute Error (**MAE**)
- Coefficient of Determination (**R^2**)

One Strength:
XGBoost achieves the highest predictive accuracy compared to other models while also providing interpretability through SHAP, making it both **accurate and explainable**.

One Weakness:

The dataset is region-specific (residential buildings from a single geographical context), which may limit the **generalisation of the model to other building types or regions**.

Overlap with Our Approach

This paper strongly overlaps with our project approach in the following ways:

- Uses **Gradient Boosting (XGBoost)**
  Directly aligned with our use of **CatBoost**, as both are boosting-based ensemble models
- Applies **feature importance analysis (SHAP)**
  Similar to our feature engineering and feature selection process
- Works on **structured tabular energy data**
  Same type of data as our project (ASHRAE building dataset)

Justification for CatBoost over Deep Learning

This paper provides indirect but strong justification for choosing CatBoost over deep learning models:

- XGBoost outperforms ANN (deep learning model) in prediction accuracy
- Boosting models perform better on **tabular datasets** compared to deep learning
- Requires **less data and computational resources** than neural networks
- Provides **better interpretability**, which is critical in energy applications

Where found in paper:
Method: Method section
Dataset: Data section
Metric: Results

- **Improving energy management practices through accurate building energy consumption prediction: analyzing the performance of LightGBM, RF, and XGBoost models with advanced optimization strategies**

Link – Improving energy management practices through accurate building energy consumption prediction: analyzing the performance of LightGBM, RF, and XGBoost models with advanced optimization strategies | Electrical Engineering | Springer Nature Link

Method, Dataset, Metric, Strength, Weakness

Method:
The study applies **ensemble machine learning models**, specifically **LightGBM, Random Forest, and XGBoost**, combined with advanced optimisation techniques such as Satin Bowerbird Optimizer (SBO) and Moth Flame Optimization for hyperparameter tuning.

Dataset Size:
The dataset consists of **real building energy consumption data collected from two buildings in Richland, Washington, covering the period 2009–2011**.
(Time-series dataset, not fixed row count → this is correct academic wording)

Key Metric:
Model performance is evaluated using:

- **R^2 (coefficient of determination)**
- **MAE (Mean Absolute Error)**
- **RMSE (Root Mean Squared Error)**

One Strength:
The study demonstrates that **gradient boosting models (LightGBM/XGBoost) achieve high prediction accuracy with optimised hyperparameters**, making them highly effective for real-world energy forecasting.

One Weakness:
The dataset is limited to **only two buildings**, which may restrict the **generalisability of the model across diverse building types and climates**.

Overlap with Our Approach

This paper has STRONG overlap with your project:

Gradient Boosting

- Uses XGBoost + LightGBM
- Same family as CatBoost

Feature-based modelling

- Uses building + environmental features
- Includes correlation analysis & feature importance

Structured tabular data

- Same type as our ASHRAE dataset

Justification for CatBoost over Deep Learning

From this paper:

- Uses tree-based boosting models instead of deep learning
- Achieves high accuracy ($R^2 > 0.91$)
- Works efficiently on tabular data
- Requires less computational cost than deep learning

Where found in paper:
Method: Methodology
Dataset: Data section
Metric: Results

- **XGBoost energy consumption prediction based on multi-system data HVAC**

Link - [2105.09945] XGBoost energy consumption prediction based on multi-system data HVAC

Method, Dataset, Metric, Strength, Weakness

Method:

The paper proposes a **machine learning framework based on XGBoost**, where features are extracted from large-scale HVAC system data. Multiple XGBoost models are trained and then combined with LightGBM predictions using a fusion strategy to improve accuracy.

Dataset Size:

The study uses **large-scale multi-system HVAC data collected from building IoT systems**, including environmental and operational variables.
Exact row count is not fixed, but it is described as **"large datasets from HVAC systems"**

Key Metric:

- **MAE (Mean Absolute Error)** → used for model fusion and evaluation
- (Also mentions MSE in related works for comparison)

One Strength:

The model effectively combines **XGBoost + feature extraction + ensemble fusion**, allowing it to capture complex relationships in HVAC energy data and improve prediction accuracy.

One Weakness:

The model relies on **large-scale IoT and sensor data**, which may limit applicability in buildings where such detailed data is not available.

Overlap with Our Approach

This paper has VERY STRONG overlap with our project:

Gradient Boosting

- Uses XGBoost (core model)
- Same family as CatBoost

Feature Engineering

- Explicitly performs feature extraction from large datasets

Structured Data

- Works on tabular HVAC + environmental data

Justification for CatBoost over Deep Learning

From this paper:

- Uses **boosting models instead of deep learning**
- Works effectively on **structured tabular data**
- Handles **large datasets efficiently**
- Uses feature-based modelling (not deep neural networks)

Where found in paper:
Method: Methodology
Dataset: Dataset description
Metric: Results

Paper	Method	Dataset Size	Key Metric	Strength	Weakness	Overlap	Justification
Gradient Boosting Comparison (LightGBM, XGBoost, CatBoost)	Ensemble boosting models	Simulated building dataset (tabular energy + environmental data; size not explicitly reported)	RMSE-type metrics	Strong comparison of boosting models	Uses single simulated dataset	Gradient boosting, tabular data, feature-based learning	Shows boosting models perform well without deep learning
CatBoost + Meta-Heuristics	CatBoost + PSO, SSA, GWO optimisation	Residential building dataset (architectural features)	RMSE, MAE	Improved accuracy via optimisation	High computational complexity	CatBoost, feature-based modelling	Shows CatBoost achieves high accuracy without deep learning
CatBoost + Bayesian Optimisation	CatBoost + Random Search + Bayesian tuning	Benchmark dataset (UCI Energy dataset, several hundred samples)	RMSE, MAE, R^2	Very high accuracy ($R^2 \approx 0.99$)	Limited real-world generalisation	CatBoost, tabular data, optimisation	Boosting models outperform complex models on structured data

Paper	Method	Dataset Size	Key Metric	Strength	Weakness	Overlap	Justification
XGBoost + SHAP (Interpretability)	XGBoost + ANN, SVM, RF comparison	3,594 residential building samples	RMSE, MAE, R^2	High accuracy + interpretability	Region-specific dataset	XGBoost, feature importance	XGBoost outperforms ANN (deep learning)
LightGBM, RF, XGBoost (Springer paper)	Ensemble boosting + optimisation	Time-series data (2 buildings, 2009–2011)	RMSE, MAE, R^2	High real-world accuracy	Limited dataset diversity	Boosting, tabular data, feature engineering	Boosting models efficient vs deep learning
XGBoost HVAC (arXiv 2105.09945)	XGBoost + feature extraction + ensemble fusion	Large-scale HVAC IoT dataset (tabular time-series)	MAE	Captures complex relationships well	Requires large IoT data	Gradient boosting, feature engineering	Efficient on structured data vs deep learning